Software Requirements Specification

Version 1.0

Chat Server Management System

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1.Introduction

Chat server management system is a software that enables users to chat with one another one-to-one or as a group all over the world. It also responsible for the payroll management of the employees. It makes life easy for the people to communicate

throughout the world.

1.1 Purpose

The goal of the system is to capture time and provide encrypted communication between users via chat, voice call and video call as a one-to-one or a group. It also provide recording voice and video calls, send images, documents, user location and text messages. This data should be further processed to backup management.

1.2 Scope

This Application works in Multiple PC's installed on multiple Computers but sharing same database by which users of different department can use it sitting at different locations simultaneously. But in future we can make the Application where the database will be hosted in order to manage the all departments which will be located in different places and by keeping domain of Application as Online.

1.3 Definitions, acronyms and abbreviations

SRS: Software Requirement Specification

CRMS: Chat Server Management System

System Operator: System administrator, data entry operator

RAM: Random Access Memory

System Administrator/Administrator: User having all the privileges to operate the CRMS.

Users: People that uses the software.

1.4 References

(a) Software Engineering by K.K. Aggarwal & Yogesh Singh, New Age Publishing House, 3rd Edition, 2008.

(b) IEEE Recommended Practice for Software Requirements Specifications – IEEE Std 8301998.

1.5 Overview

This system is developed for communication via text messages, video call, voice call, Group chat, chat with friend, Group calls. It also responsible for the tracking the users fast and encrypted communication. It makes life easy for the people throughout the world.

The rest of the SRS document contains the purpose and features of the software, the interfaces of the software, what the software will do, the constraints under which it must operate and how the software will react to external stimuli. This document is intended both for the end users and the developers of the software. The sections of the SRS document that lies ahead are: Overall Description, Specific Requirements and Supporting Information.

2. Overall Description

The CRMS is a software developed for communication via text messages, video call, voice call, Group chat, chat with friend, Group calls. It also responsible for the tracking the users fast and encrypted communication. It makes life easy for the people throughout the world.

The administrator will have to maintain the following information:

- User Details
- User History
- User chat
- Users Group and friend list

The administrator will perform the following functions:

- Notification
- Login and logout Management

The User requires the following information :

- Login Id
- Login Password

2.1 Product Perspective

The CRMS shall be developed using Chat server architecture and will be compatible with

Microsoft Windows Operating System. The front end of the system will be developed using HTML and CSS, and the back-end will be developed using PHP and MySQL. It provides simple database rather than complex ones for high requirements and it provides good and easy graphical user interface to both new as well as experienced user of the computer.

2.1.1 System Interfaces

None

2.1.2 User Interfaces

The CRMS will have the following user-friendly and menu-driven interfaces:

(a) Login: to allow the entry of only authorized users through valid login ID and password.
(b) Add Friend: Allow user to add new friends in their database.
(c) Remove Friend: Allow user to remove or delete friends from their database.
(d) Block Friend: Allow user to block a friend on the system.
(e) Group Chat: Allow user to create a group or add in a group.
(f) Log out: Allow user to log out from the system.

2.1.3 Hardware Interfaces

(a) Screen Resolution of at least 640 * 480 or above.
(b) Support for printer.
(c) Computer systems will be in the networked environment as it is a multi-user system.

2.1.4 Software Interfaces

(a) MS-Windows Operating System (XP/Vista/8/10) (b) HTML and CSS for designing front end.
(c) PHP and MySQL for designing the back end.

2.1.5 Communication Interfaces

In the CRMS, communication is via local area network (LAN) and it is a web-enabled system.

2.1.6 Memory Constraints

At least 2 GB RAM and 200 MB space of hard disk will be required to run the software.

2.1.7 Operations

None

2.1.8 Site Adaptation Requirements

The terminal at the client side will have to support the hardware and software interfaces specified in sections 2.1.3 and 2.1.4, respectively.

2.2 Product Functions

The CRMS will allow access only to the authorized users with specific roles (system administrator, DEO and Users). Depending on the user's role he/she will be able to access only specific modules of the system.

A summary of major functions that the LMS shall perform includes:

- A login facility for enabling only authorized access to the system.
- The system administrator will be able to add, modify, delete or view chat between the users.
- The system administrator will be able to verify the login details given by user.

- The system administrator will be able to keep track of all the user login history, users information and chat history.
- The system administrator will be able to generate various reports from the CRMS.

2.3 User Characteristics

Qualification: At least matriculation and comfortable with English.

Technical Experience: Basic knowledge of computers.

2.4 Constraints

- There will be at most 3 administrators and at least 1.
- The delete operation is available to the User. To reduce the complexity of the system, there is no check on delete operation. Hence, the User should be very careful before deletion of any record and he/she will be responsible for data consistency.
- The system must be designed using PHP and MySQL.
- The system must be secure against XSS attacks and SQL injections.

2.5 Assumptions and Dependencies

- Contact list of user and have account on chat server are show to add new friends.
- To add a new friend both users have permission are necessary.
- The login ID and password must be created by the User and communicated to the concerned user confidentially to avoid unauthorized access to the system.

2.6 Apportioning of Requirements

The CRMS currently lacks the functionality of pause a account for some time. These requirements will be incorporated in the future version of the software system.

3. Specific Requirements

This section contains the software requirements in detail along with the various forms to be developed.

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

(a) Screen Resolution of at least 640 * 480 or above.
(b) Support for printer.
(c) Computer systems will be in the networked environment as it is a multi-user system.

3.1.3 Software Interfaces

(a) MS-Windows Operating System (XP/Vista/8/10) (b) HTML and CSS for designing front end.
(c) PHP and MySQL for designing the back end.

3.1.4 Communication Interfaces

In the CRMS, communication is via local area network (LAN) and it is a web-enabled system.

3.2 Functional Requirements

3.2.1 Register

A. Use Case Description

Introduction: This use case allows the user to register the system.
Actors: User Admin
Precondition: The user must have a phone number or their email.
Postcondition: If the use case is successful,the user is granted access to the system.
Event Flow: Basic Flow 1. The user installs the chat application. 2. The user enters his/her phone number or email id and signs up . 3. The users login id i.e phone number and password are saved with the system's database.
Alternative Flow: 1. If the user installs the app but does not register, and tries to login then an error message will be flagged saying please sign up first.And use case returns to the beginning of the basic flow. 2. This allows the user to exit the server any time during the use case.The use case ends.
Special requirements: phone number
Associated use cases: none

B. Validity Checks

i. Every user will have a unique login ID.
ii. Login ID cannot be blank.
iii. Login ID can only have 4 to 15 characters.
iv. Login ID will not accept special characters and blank spaces.
v. Password cannot be blank.
vi. Length of password can only be 4 to 15 digits.
vii. Password will not accept blank spaces.

C. Sequencing Information

None

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.2 Login

A. Use Case Description

Introduction: This use case allows the user to access the system.
Actors: Administrator User
Precondition: The user must have their login id and password with them.
Postcondition: If the use case is successful ,the user is granted access to the system.
Event Flow: Basic Flow 1.The user enters his/her login id password. 2.The user's login id password is matched with the system's database. 3.The user's credentials match and he is granted access to the system.
Alternative Flow: If the system does not validate the user's login id or password due to entering wrong credentials then an error message is flagged and the use case returns to the beginning of the basic flow.
Special requirements: None
Associated use cases: none

B. Validity Checks

iv. Every user will have a unique login ID.
v. Login ID cannot be blank.
vi. Login ID can only have 4 to 15 characters.
viii. Login ID will not accept special characters and blank spaces.
ix. Password cannot be blank.
x. Length of password can only be 4 to 15 digits.
xi. Password will not accept blank spaces.

C. Sequencing Information

None

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be

prompted to the user for doing the needful.

3.2.3 Logout

A. Use Case Description

Introduction: This use case allows the user to logout the system.
Actors: User
Precondition: The user must be logged in to the system.
Postcondition: If the use case is successful then the user can log out of the system.
Event Flow: Basic Flow User is done using the web application 1. The user clicks on the logout button 2. The system logs the user out and invalidates the cookie/session 3. The system redirects to the default home page The user logs out of the computer.
Alternative Flow: None
Special requirements: none
Associated use cases: login

3.2.4 Friend list

A. Use Case Description

Introduction: This allows users to check their friend list.
Actors: User
Precondition: User must be logged into the system.
Postcondition: If the use case is successful then the user can check their friend list.
Event Flow: Basic Flow 1.User enters the required details. 2.User clicks on the friends button.
Alternative Flow: If the user does not have any friends then he/she is flagged with a message no friends yet add friends and the use case is sent to find a friend.
Special requirements: none

Associated use cases: add a friend

B. Validity Checks

i. Only the User will be authorized to access the Friend List Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.5 Find Friend

A. Use Case Description

Introduction: This allows users to find a friend on the system.
Actors: user
Precondition:user must be logged into the system.
Postcondition: if the use case is successful then the user can find their friend.
Event Flow: Basic Flow 1.User search for his/her friend in the search tab. 2.User enter his friend's name or phone number. 3.User clicks on the friend's profile.
Alternative Flow: If the user enters a name which is not in the system's database then an error message is flagged saying no results found and the use case returns to the beginning of the basic flow.
Special requirements: none
Associated use cases: login

B. Validity Checks

i. Only the User will be authorized to access the Find Friend Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.6 Add Friend

A. Use Case Description

Introduction: This use case allows the user to add a friend on the system.
Actors: User
Precondition: The user must be logged into the system before adding a friend.
Postcondition: If the use case is successful then the user is connected to his/her friend on the system.
Event Flow: Basic Flow 1.The user selects another user and sends an invitation by clicking the request button. 2.The system will send a request message to the other user and show a message to the first user to inform that the action has been done.
Alternative Flow: If the user has already sent an invitation to the other user, the system will inform the user.
Special requirements: none
Associated use cases: login

B. Validity Checks

i. Only the User will be authorized to access the Add Friend module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.7 Remove Friend

A. Use Case Description

Introduction: This use case allows the user to remove a friend on the system.

Actors: Users
Precondition: The user must be friends with the user he/she wants to remove.
Postcondition: If the use case is successful then the user can remove his/her friend on the system.
Event Flow: Basic Flow 1.The user opens his friend profile. 2.The user selects the option-remove as friend on his/her friend's profile.
Alternative Flow: If the user has already removed his friend, the system will inform the user.
Special requirements: none
Associated use cases: add friend

B. Validity Checks

i. Only the User will be authorized to access the Remove Friend Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.8 Block Friend

A. Use Case Description

Introduction: This use case allows the user to block a friend on the system.
Actors: Users
Precondition: The user must enter the name of the user he/she wants to block.
Postcondition: If the use case is successful then the user can block his/her friend on the system.
Event Flow: Basic Flow 1.The user opens his friend profile. 2.The user selects the option-block on his/her friend's profile.
Alternative Flow: none

Special requirements: none
Associated use cases: login

B. Validity Checks

i. Only the User will be authorized to access the Block Friend Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.9 Group Chat

A. Use Case Description

Introduction: This use case allows the user to chat with many friends at a time i.e group chat on the system.
Actors: Users
Precondition: The user must be logged in to the system and must be connected to his/her friends.
Postcondition: If the use case is successful ,then the user sends the message on the system.
Event Flow: Basic Flow 1.The user ,when viewing their friends and acquaintances list can select the option to create a new group. 2.The user sets the group name. 3.The user select which of their friends and acquaintances will be added to the group before confirming their selection. 4.A group chat is created. Anyone in the group can send the message and everyone in the group can receive the message.
Alternative Flow: If the user tries to add friends more than the limit then an error message is flagged saying you cannot add more friends and the use case returns to the basic flow.
Special requirements: none
Associated use cases: Contact

B. Validity Checks

i. Only the Group Admin and User will be authorized to access the Group Chat Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be prompted to the user for doing the needful.

3.2.10 Feedback

A. Use Case Description

Introduction: This use case allows feedback about the system.
Actors: User,Admin
Precondition: User and admin must be active.
Postcondition: If the use case is successful ,then the user gives feedback to the server.
Event Flow: Basic Flow 1. User selects "Provide Feedback" 2.System looks up central user support phone numbers. 3.System displays phone numbers to call and gives the user the option to send email immediately 4.User selects email option 5.System looks up the email address of customer service and passes it to the browser. 6.System launches email browser client. 7.User enters message, presses "Send" 8.Browser mail client sends mail.
Alternative Flow: If an intruder is detected to open the account of the user ,then the administrator sends a recovery code on the user's phone number.
Special requirements: none
Associated use cases: none

B. Validity Checks

i. Only the User will be authorized to access the Feedback Module.

C. Sequencing Information

User must be Register in the system.

D. Error Handling

If any of the validation flows does not hold true, appropriate error message will be

prompted to the user for doing the needful.

3.3 Performance Requirements

(a) Should support at least 10 terminals.

(b) Should support at least 10 users simultaneously.

(c) Should run on 1.3 GHz, 2GB RAM machine. (d) Responses should be within 2 seconds.

3.4 Design Constraints

None

3.5 Software System Attributes

Usability

The application will be user-friendly and easy to operate, and the functions will be easily understandable.

Reliability

The applications will be available to the students throughout the registration period and have a high degree of fault tolerance.

Security

The application will be One-time-password protected. Users will have to enter correct Mobile Number and One-time-password to access the application.

Maintainability

The application will be designed in a maintainable manner. It will be easy to

incorporate new requirements in the individual modules.

Portability

The application will be easily portable on any windows-based system that has SQL server installed.

Other Requirements

None

TEST CASES

1. LOGIN

Use Case Scenario Matrix		
Scenario Number and Description	**Originating Flow**	**Alternative Flow**
Scenario 1: Login Basic Flow	Basic Flow	
Scenario 2: Invalid Login-Id or Password	Basic Flow	Alternative Flow 1
Scenario 3: User Exits	Basic Flow	Alternative Flow 2

Test Case Matrix						
Test Case ID	**Scenario Number and Description**	**Input 1: Login ID**	**Input 2: Password**	**Expected Output**	**Actual Output**	**Remarks (if any)**
TC1	Scenario 1: Login Basic Flow	Valid	Valid	User is allowed to login	User is allowed to login	-
TC2	Scenario 2: Login Alternative Flow: Invalid Login-Id or Password	Invalid	Valid/Invalid	Invalid Login-Id	Invalid Login-Id	Login-Id does not exist in database or not in valid format.
TC3	Scenario 2: Login Alternative Flow: Invalid Login-Id or Password	Valid	Invalid	Invalid Password	Invalid Password	Password does not exist in database or not in valid format.
TC4	Scenario 3: Login Alternative Flow: User Exits	Valid/Invalid	Valid/Invalid	User comes out of the System.	User comes out of the System.	-

Test Case ID	Scenario Number and Description	Input 1: Login ID	Input 2: Password	Expected Output	Actual Output	Remarks (If any)
TC1	Scenario 1: Login Basic Flow	463471423	Abc123	User is allowed to login	User is allowed to login	-
TC2	Scenario 2: Login Alternative Flow: Invalid Login-Id or Password	787257234	Abc1234	Invalid Login-Id	Invalid Login-Id	Login-Id does not exist in database or not in valid format.
TC3	Scenario 2: Login Alternative Flow: Invalid Login-Id or Password	893698345	Abc124	Invalid Password	Invalid Password	Password does not exist in database or not in valid format.
TC4	Scenario 3: Login Alternative Flow: User Exits	*	*	User comes out of the system	User comes out of the system	-

2. Contact Form

Use Case Scenario Matrix		
Scenario Number and Description	**Originating Flow**	**Alternative Flow**
Scenario 1: Friend list	Basic Flow	

Scenario 2: User Exit	Basic Flow	Alternative Flow 1
Scenario 3: Add Friend	Basic Flow	
Scenario 4: User Exit	Basic Flow	Alternative Flow 2
Scenario 5: Remove Friend	Basic Flow	
Scenario 6: User Exit	Basic flow	Alternative Flow 3
Scenario 7: Block	Basic Flow	
Scenario 8: User Exit	Basic Flow	Alternative Flow 4
Scenario 9: Group chat	Basic Flow	
Scenario 10:Feedback	Basic Flow	Alternative Flow 5

Test Case Matrix					
Test Case ID	**Scenario Number and Description**	**Input 1: User Id**	**Expected Output**	**Actual Output**	**Remarks (if any)**
TC1	Scenario 1: Friend list	3463452524	Show the friend list	Friend list display.	-
TC2	Scenario 1: Friend list	2398729364	Show the friend list	Friend list not display.	Wrong User Id
TC3	Scenario 2: Friend List Alternative Flow: User Exit		User comes out of the system.	User comes out of the system.	-

Test Case Matrix					
Test Case ID	**Scenario Number and Description**	**Input 1: User Id**	**Expected Output**	**Actual Output**	**Remarks (if any)**

TC1	Scenario 3: Add Friend	6463434524	Friend is added.	Friend is added.	-
TC2	Scenario 3: Friend list	534535244	Friend is added.	Adding a friend failed.	Wrong User Id
TC3	Scenario 4: Friend List Alternative Flow: User Exit		User comes out of the system.	User comes out of the system.	-

Test Case Matrix					
Test Case ID	**Scenario Number and Description**	**Input 1: User Id**	**Expected Output**	**Actual Output**	**Remarks (if any)**
TC1	Scenario 5: Remove Friend	3463452524	Friend is Removed.	Friend is Removed.	-
TC2	Scenario 5: Friend list	2398729364	Friend is Removed.	Friend not Found.	Wrong User Id
TC3	Scenario 6: Remove Friend Alternative Flow: User Exit		User comes out of the system.	User comes out of the system.	-

Test Case Matrix					
Test Case ID	**Scenario Number and Description**	**Input 1: User Id**	**Expected Output**	**Actual Output**	**Remarks (if any)**
TC1	Scenario 7: Block Friend	3463452524	Friend is blocked.	Friend is blocked.	-

TC2	Scenario 7: Friend list	2398729364	Friend is blocked.	Friend is not Found.	Wrong User Id
TC3	Scenario 8: Friend List Alternative Flow: User Exit		User comes out of the system.	User comes out of the system.	-

Test Case Matrix					
Test Case ID	**Scenario Number and Description**	**Input 1: Group Name**	**Expected Output**	**Actual Output**	**Remarks (if any)**
TC1	Scenario 9: Group Chat	ABCD	Group Found	Group chat is display	-
TC2	Scenario 9: Group Chat	Abcdef	Group Found	Group not Found.	Wrong Group Name
TC3	Scenario 10: Friend List Alternative Flow: Feedback		User comes out of the system.	User comes out of the system.	-